Total No. of Questions : 8]

SEAT No. : ____

PA-1663

[Total No. of Pages : 2

[5927]-347

B.E. (Computer Engg.)

CYBER SECURITY AND DIGITAL FORENSICS

(2019 Pattern) (Elective-III) (Semester-VII) (410244(C))

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) ***Answer Q1 or Q2, Q3, or Q4, Q5 or Q6, Q7 or Q8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data if necessary.***
5) ***Use of scientific calculator is permitted.***

Q1) a) What is computer forensics? What is the use of Computer Forensics in Law Enforcement? What are different computer forensics schemes? **[9]**

b) Explain in detail different Computer Forensics Services. **[9]**

OR

Q2) a) Explain in details Computer Forensics Assistance to Human Resources. **[9]**

b) What are the benefits of professional forensics methodology? What are steps taken by computer forensics specialist? **[9]**

Q3) a) What are evidences? What are the simple reasons to collect evidences? What are different options for collecting evidences? **[8]**

b) What is chain of custody? Explain the process of chain of custody. **[9]**

OR

Q4) a) What is the general procedure for evidence collection? **[8]**

b) What are general computer evidence processing steps? Explain in detail. **[9]**

Q5) a) Explain how to perform remote and live acquisitions with an appropriate example. **[9]**

b) What are the different approaches for validating forensic data? **[9]**

OR

P.T.O.

CEGP013091 49.248.216.238 09/01/2023 13:55:36 static-238

Study material provided by: V. B. L

Join Community by clicking below links

Telegram Channel

https://t.me/SPPU_TE_BE_COMP

(for all engineering Resources)

SPPU Engineering & Technical UPDATES
WhatsApp channel

WhatsApp Channel

(for all Engg & tech updates)

https://whatsapp.com/channel/0029ValjFriICVfpcV9HFc3b

@SPPU_ENGINEERING_UPDATE

Insta Page

(for all Engg & tech updates)

https://www.instagram.com/sppu_engineering_update

Q6) a) Brief about the approaches for seizing digital evidence at the crime scene. **[9]**

b) Give in detail the different techniques to hide data in digital forensics. **[9]**

Q7) a) Explain types of digital forensics tools. Also explain the task performed by these tools. **[8]**

b) State the features of any five computer forensics software tools. **[9]**

OR

Q8) a) Explain the role of client and server in email and some of the tools for email forensics. **[8]**

b) Explain the process for validating and testing forensics software. **[9]**

● ● ●

CEGP013091 49.248.216.238 09/01/2023 13:55:36 static-238

Total No. of Questions : 8]

SEAT No. :

P550

[Total No. of Pages : 2

[6004]-485

B.E. (Computer Engineering)

CYBER SECURITY AND DIGITAL FORENSICS

(2019 Pattern) (Semester - VII) (Elective - III) (410244C)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Use of Calculator is allowed.***
5) ***Assume suitable data if necessary.***

Q1) a) Explain in brief computer forensic services. Write the applications of digital forensics in military. **[8]**

b) What is the significance of data recovery and backup? Explain various data recovery solutions. **[9]**

OR

Q2) a) What are the various business oriented digital forensic techniques? **[8]**

b) How does computer forensics help in law enforcement? **[9]**

Q3) a) Discuss the various legal aspects of collecting and storing digital evidence. **[9]**

b) What are different computer evidence processing steps? **[9]**

OR

Q4) a) What is chain custody? How we can control the contamination of digital evidence? **[9]**

b) What are various methods of collecting digital evidence ?Enlist the various digital collection steps. **[9]**

Q5) a) Explain different approaches for validating forensic data. **[9]**

b) Explain the approaches for seizing digital evidence at the crime scene. **[8]**

OR

P.T.O.

49.248.216.238 24/06/2023 14:42:48 static-238 CEGP013091

Q6) a) Explain the process of identifying digital evidence in computer forensics. **[9]**

b) Explain Network forensics and order of volatility for computer system. **[8]**

Q7) a) State the features of any five computer forensic software tools. **[9]**

b) Write short notes on **[9]**

i) Task performed by digital forensic tool

ii) Tools for email forensics

iii) Techniques for email forensic investigation

OR

Q8) a) State the features of any five computer forensic hardware tools. **[9]**

b) Write short notes on **[9]**

i) Role of client and server in email

ii) Investigating Email crimes and investigations

iii) NIST standards for forensic technologies



CEGP013091 49.248.216.238 24/06/2023 14:42:48 static-238

Total No. of Questions : 8]

SEAT No. : ____

P-6556

[Total No. of Pages : 2

[6181]-106

B.E. (Computer Engineering)

CYBER SECURITY AND DIGITAL FORENSICS

(2019 Pattern) (Semester - VII) (410244 C) (Elective - III)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates :

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data, if necessary.***

Q1) a) What are the typical steps followed by computer forensics specialists in an investigation? Explain any 2 in detail. **[9]**

b) In what ways can business benefit from computer forensics technology? Explain in detail. **[9]**

OR

Q2) a) What kind of digital evidences can be collected in computer forensics? Explain in detail. **[9]**

b) Why is data backup & recovery important in computer forensics? **[9]**

Q3) a) What is the primary purpose of collecting evidence in digital forensics? Explain in detail. **[9]**

b) What are the typical steps involved in the collection of digital evidences? **[8]**

OR

Q4) a) Explain the different types of digital evidence that can be collected in computer forensics? **[8]**

b) What method & techniques are commonly used to verify & authenticate computer images. explain any two in detail? **[9]**

P.T.O.

CEGP013091 49.248.216.238 07/12/2023 13:45:30 static-238

Q5) a) How do investigators determine which data is relevant to collect & analyze in digital forensics investigation? **[8]**

b) What is the honeynet project, how does it contribute to network forensics? **[9]**

OR

Q6) a) Why is data validation crucial in digital forensics & what methods are commelily used for data validation? **[8]**

b) Describe the process of seizing digital evidence at a crime or incident scene? **[9]**

Q7) a) How does email play a significant role in digital investigations? What types of information can be obtain from email Header that may be relevant in investigations? **[9]**

b) What factors should be considered when evaluating computer forensics tool needs for an investigations. Explain any two in detail? **[9]**

OR

Q8) a) What is function e-mail server, how does it store & manage e-mail data? **[9]**

b) Write short note (any one) : **[9]**

i) e-mail forensics tools.

ii) Computer forensics hardware tools

▽▽▽▽

CEGP013091 49.248.216.238 07/12/2023 13:45:30 static-238

Total No. of Questions : 8] **SEAT No. :**

PB2248 **[Total No. of Pages : 2**

[6263]-86
B.E. (Computer Engineering)
CYBER SECURITY AND DIGITAL FORENSICS
(2019 Pattern) (Semester - VII) (410244 C) (Elective - III)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right side indicate full marks.***
4) ***Assume suitable data if necessary.***

Q1) a) What are the typical services offered by computer forensics professionals? (explain any two) **[9]**

b) What specific technologies are utilized in the field of business computer forensics? (Describe any two) **[9]**

OR

Q2) a) How does computer forensics technology vary cross different sectors like miltary law enforcement & business? **[9]**

b) What are the key components of a data recovery solutions in computer forensics? Explain in detail. **[9]**

Q3) a) What are various options available for collecting digital evidences? Explain in detail? **[8]**

b) Explain the essential steps in processing digital evidence from the crime scene? **[9]**

OR

Q4) a) What is the volatile evidence in the context of computer forensics, and why is it important to collect in quickly? **[8]**

b) What are the hypical steps involve in the collection of digital evidence? **[9]**

Q5) a) What are some common data hiding techniques? Explain any one in detail? **[8]**

b) What is the honey net project, and how does it contribute to network forensics? **[9]**

OR

P.T.O.

CEGP013091 49.248.216.238 18/05/2024 13:59:26 static-238

Q6) a) What precautions should investigators take to prevent data alteration or loss during the seicure process? Explain any one in detail? **[8]**

b) What are the challenges and best practices associated with performing remove acqisitions? **[9]**

Q7) a) Whar are the common examples of email crime & violations that may necessitate investigation? Explain any one in detail? **[9]**

b) Explain any software tool using in computer forensics investigation and its respective purpose? **[9]**

OR

Q8) Write short note (any two) **[18]**

a) Computer forensics hardware tools

b) Validating & testing forensics software

c) e-mail investigation

CEGP013091 49.248.216.238 18/05/2024 13:59:26 static-238

Total No. of Questions : 8]

SEAT No. :

PC2373

[Total No. of Pages : 2

[6354]-490

B.E. (Computer Engineering)
CYBER SECURITY AND DIGITAL FORENSICS
(Elective -III) (2019 Pattern) (Semester-VII) (410244 C)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4,Q.5 or Q.6, Q.7, or Q.8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data, if necessary.***

Q1) a) What are the typical steps followed by computer forensics specialists in an investigation? Explain any 2 in detail. **[8]**

b) In what ways can business benefit from computer forensics technology explain in details? **[9]**

OR

Q2) a) Explain in details different computer forensics services. **[9]**

b) Why is data backup & recovery important in computer forensics? **[8]**

Q3) a) What are the typical steps involved in the collection of digital evidence?**[9]**

b) What are the different approaches for validating forensic data? **[9]**

OR

Q4) a) Why collect evidence? collection options in digital evidence. explain in details? **[9]**

b) Discuss the various legal aspects of collecting and storing digital evidence. **[9]**

Q5) a) What are same common network tools, used in network forensics. **[9]**

b) Describe the process of seizing digital evidence at a crime or incident scene? **[8]**

OR

P.T.O.

Q6) a) What is the honeynet project, how does it contribute to network forensics? **[9]**

b) Give in detail the different techniques to hide data in digital forensics? **[8]**

Q7) a) Write short note on. **[9]**

i) Tools for email forensics

ii) Computer forensics hardware tools

b) Explain the process for validating and testing forensics software. **[9]**

OR

Q8) a) What is function email server show does it store of mange e-mail data? **[9]**

b) Write short note on. **[9]**

i) Computer forensics software tools

ii) E-mail Investigations

■■■■